For instance, if we are looking for the shortest path in a graph, we can define an equivalent decision problem by setting a threshold T and checking if there exist a path with length at most T: by solving this problem for different values of T, and by choosing these values using binary search, we can find a solution to the optimization problem by using an algorithm for the decision problem[1].

F.2 Las Vegas Algorithms

The class of so-called “Las Vegas algorithms” includes all randomized algorithms that:

1. Always gives as output the correct solution to the problem (or returns a failure).
2. Whose execution can take a random (though finite) amount of resources, which can't be predicted based on input.

The most notable example of a Las Vegas algorithms is probably randomized quicksort[2]: it always produces the correct solution, but the pivot is chosen randomly at every recursive step, and execution time swirls between $O(n*log\ n)$ and $O(n^2)$ depending on the choices made.

F.3 Monte Carlo Algorithms

We classify algorithms as Monte Carlo methods, instead, when the output of the algorithm can sometimes be incorrect. The probability of an incorrect answer is usually a